

MITRE D3FEND Overview

D3FEND is a knowledge graph of cybersecurity countermeasures.

This document is designed as clean, selectable text for a RAG knowledge base.

What MITRE D3FEND Is

MITRE D3FEND is a cybersecurity knowledge base and knowledge graph that describes defensive countermeasure techniques. It standardizes terminology for defensive technology functions and relates defensive techniques to digital artifacts, system components, and adversary behavior.

Why It Matters

Security teams often describe defenses using product names or inconsistent terms. D3FEND focuses on what a defensive capability does rather than which vendor supplies it.

Core Defensive Purposes

Purpose	Description
Model	Understand systems, assets, relationships, risks, weaknesses, threats, and defensive requirements.
Harden	Change configurations, access controls, software, data, or design to reduce attack opportunities.
Detect	Observe and analyze events, behaviors, artifacts, and state changes that may indicate malicious activity.
Isolate	Restrict communication, execution, privileges, or access to limit adversary movement and impact.

Deceive	Use decoys or deceptive environments to mislead adversaries and reveal unauthorized interaction.
Evict	Remove malicious processes, accounts, persistence, files, access, or other adversary footholds.
Restore	Return systems, data, identities, and services to a trusted operational state.

Example Defensive Techniques

- Authentication and authorization controls
- Network monitoring and traffic analysis
- Operating-system and process monitoring
- File and content analysis
- Credential protection and revocation
- Decoy environments and honeypots
- Segmentation and isolation
- Recovery and restoration

Relationship to MITRE ATT&CK

MITRE ATT&CK describes what adversaries do. D3FEND describes defensive countermeasure techniques. D3FEND helps teams move from “What is the adversary doing?” to “What defensive functions can detect, disrupt, contain, or recover from that behavior?”

SOC Workflow

1. Identify suspicious behavior and map it to an ATT&CK technique when appropriate.
2. Determine which artifacts are involved, such as credentials, processes, files, traffic, accounts, or configuration data.
3. Use D3FEND terminology to identify relevant countermeasure techniques.
4. Review existing controls and available telemetry.
5. Recommend detection, hardening, isolation, eviction, deception, or restoration actions.

6. Document the chosen controls and the evidence they address.

Phishing Example

For a phishing incident, ATT&CK can classify spearphishing or credential-access behavior. D3FEND can support recommendations such as email-content analysis, URL analysis, domain checks, authentication controls, credential revocation, account monitoring, network isolation, and system restoration.

Suggested RAG Questions

- What is MITRE D3FEND?
- How does D3FEND relate to MITRE ATT&CK?
- Which defensive techniques help detect credential theft?
- What is a decoy environment?
- How can D3FEND support phishing incident response?

References

MITRE D3FEND: <https://d3fend.mitre.org/>

D3FEND FAQ: <https://d3fend.mitre.org/faq/>

MITRE ATT&CK: <https://attack.mitre.org/>